

The ability to sense the environment is heterogeneously distributed in cell populations

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

January 4, 2024 (this version)

Reviewed preprint version 1

June 22, 2023

Sent for peer review

March 30, 2023

Posted to preprint server

March 13, 2023

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Abstract

Channel capacity of signaling networks quantifies their fidelity in sensing extracellular inputs. Low estimates of channel capacities for several mammalian signaling networks suggest that cells can barely detect the presence/absence of environmental signals. However, given the extensive heterogeneity and temporal stability of cell state variables, we hypothesize that the sensing ability itself may depend on the state of the cells. In this work, we present an information theoretic framework to quantify the distribution of sensing abilities from single cell data. Using data on two mammalian pathways, we show that sensing abilities are widely distributed in the population and most cells achieve better resolution of inputs compared to an “average cell”. We verify these predictions using live cell imaging data on the IGFR/FoxO pathway. Importantly, we identify cell state variables that correlate with cells’ sensing abilities. This information theoretic framework will significantly improve our understanding of how cells sense in their environment.

eLife assessment

In this **valuable** paper, the authors use an existing theoretical framework relying on information theory and maximum entropy inference in order to quantify how much information single cells can carry, taking into account their internal state. They reanalyze experimental data in this light. Despite some limitations of the data, the study **convincingly** highlights the difference between single-cell and population channel capacities. This result should be of interest to the quantitative biology community, as it contributes to explaining why channel capacities are apparently low in cells.

Introduction

In cell populations, there is a significant overlap in responses to environmental stimuli of differing strengths. This raises a fundamental question¹: do signaling networks in cells relay accurate information about their environment to take appropriate action? And if not, where along the signal transduction pathway is the information lost²³? Mutual information (MI) quantifies

the information content in an intracellular output about extracellular inputs. For an input u (e.g., concentration of a ligand) and an output x (e.g., intracellular species concentration^{4,6} or a cellular phenotype^{7,8}), the MI is defined as⁹

$$I = \sum_{x,u} p(x|u)p(u) \log_2 \frac{p(x|u)}{\sum_{u'} p(x|u')p(u')}. \quad (1)$$

Experimental single cell methods such as flow cytometry¹⁰, immunofluorescence¹⁰, mass spectrometry¹¹, or live cell imaging¹² allow us to estimate response histograms $p(x|u)$ across several inputs. Using these distributions, we can estimate the maximum of the MI (the channel capacity, CC) by optimizing Eq. 1 over input distributions $p(u)$. The CC quantifies fidelity of signal transduction^{1,2}. For example, a CC of 1 bit implies that the cells can at best distinguish between two input levels (e.g. presence versus absence), with higher CCs indicating that cells can resolve multiple input states. Importantly, CC can be used to identify bottlenecks in signaling^{2,13}.

The CC has been estimated for input-output relationships in several mammalian signaling networks^{4,6,14-16}. When the output is defined as levels of a single protein a fixed time, the CC was found to be surprisingly low, $\sim 1 - 1.5$ bits. These estimates have been improved by considering multidimensional outputs⁵ or time varying inputs¹⁷. While these modifications led to somewhat higher CC estimates, the overall conclusion that cells know little about their environment remains well established. In contrast, significantly higher CC estimates have been found when the output at the level of cell population averages are considered⁶, suggesting that the only way to overcome low sensing fidelity is population average response.

These previous calculations estimated one channel capacity for all cells in a population, implicitly assuming that individual cells have similar sensing abilities. However, we now know that cells exhibit extensive heterogeneity in cell state variables^{18,19} such as abundances of key signaling proteins^{20,21}, mRNA abundances²², and chromatin conformation²³ and accessibility²⁴. Notably, the time scale of fluctuations in these variables can be significantly slower than relevant signaling time scales^{25,26}, sometimes extending across multiple generations²⁷. Heterogeneity in cell state variables leads to a heterogeneity in response to extracellular cues, including chemotherapeutic drugs^{25,26,28}, mitogens²⁹, hormones³⁰, chemotactic signals^{7,27,31,32}, and other electrical and chemical stimuli^{33,34}. Therefore, we hypothesize that the ability to sense the environment varies from cell-to-cell in populations, in a cell state dependent manner.

There is no conceptual framework to estimate the variation in sensing abilities in cell populations and its dependence on cell state variables. To that end, we introduce an information theoretic quantity CeeMI: **C**ell state dependent **M**utual **I**nformation. We show that using typically collected single cell data and computational models of signaling networks, we can estimate the distribution $p_{\text{CeeMI}}(I)$ of single cell signaling fidelities (single cell mutual information values). We also show that we can identify cell state variables that make some cells better and others worse at sensing their environment.

Using an illustrative example, we show that in heterogeneous cell populations, estimates of mutual information that average over cell states can be significantly lower than the mutual information of signaling networks in typical cells. Next, using previously collected experimental data, we estimate $p_{\text{CeeMI}}(I)$ for two important mammalian signaling pathways³⁵; the Epidermal growth factor (EGF)/EGF receptor pathway and the Insulin like growth factor (IGF)/Forkhead Box protein O (FoxO) pathway, we show that while the cell state agnostic CC estimates for both pathways are ~ 1 bit, most individual cells are predicted to be significantly better at resolving different inputs. Using live cell imaging data for the IGF/FoxO pathway, we show that our estimate of variability in sensing abilities matches closely with experimental estimates. Importantly, for this pathway, we

also verify our prediction that specific cell state variables dictate cells' sensing abilities. Finally, using a simple receptor/ligand model, we show how the time scales cell state dynamics affects cells' individuality and therefore sensing abilities. We believe that CeeMI will be an important tool to explore variability in cellular sensing abilities and in identifying what makes some cells better and others worse in sensing their environment.

Results

Conditional mutual information models single cells as cell state dependent channels

Consider a cell population where cells are characterized by state variables θ . These include abundances of signaling proteins and enzymes, epigenetic states, etc. We assume that cell states are temporally stable, that is, θ remains constant over a time scale that is longer than typical fluctuations in environmental inputs. Later, using a toy simple model, we explicitly study the effect of cell state dynamics on cells' ability to sense their environment.

We assume that cell state variables are distributed in the population according to a distribution $p(\theta)$. If x denotes an output of choice (e.g., phosphorylation levels of one or more protein(s) at one or more time point(s)) and u denotes the input (e.g., ligand concentration), the experimentally measured response distribution $p(x|u)$ can be decomposed as:

$$p(x|u) = \int p(x|u, \theta) p(\theta) d\theta. \quad (2)$$

where $p(x|u, \theta)$ is the distribution of the output x conditioned on the input u and cell state variables θ . We note that in most cases, the same cell cannot be probed multiple times. Consequently, $p(x|u, \theta)$ may not be experimentally accessible. However, it is conceptually well defined and mathematically accessible when interactions amongst molecules are specified³⁶.

Using $p(x|u, \theta)$, we can define the cell state dependent mutual information $I(\theta)$ for a fixed input distribution $p(u)$ analogously to Eq. 1:

$$I(\theta) = \sum_{x,u} p(x|u, \theta) p(u) \log_2 \frac{p(x|u, \theta)}{\sum_{u'} p(x|u', \theta) p(u')}. \quad (3)$$

$I(\theta)$ quantifies individual cells' ability to sense their environment as a function of the cell state parameters θ . The distribution $p_{\text{CeeMI}}(I)$ of single cell sensing abilities is

$$p_{\text{CeeMI}}(I) = \int p(\theta) \delta(I - I(\theta)) d\theta \quad (4)$$

where $\delta(\cdot)$ is the Dirac delta function. We can also compute the joint distribution between the single cell mutual information and any cell state variable of interest χ (e.g., abundance of cell surface receptors):

$$p_{\text{CeeMI}}(I, \chi) = \int p(\theta) \delta(I - I(\theta)) \delta(\chi - \chi(\theta)) d\theta \quad (5)$$

where $\chi(\theta)$ is the value of the biochemical parameter when cell state variables are fixed at θ . The distributions in Eq. 5 quantify the interdependency between a cell's signaling fidelity $I(\theta)$ and cell specific biochemical parameters $\chi(\theta)$. As we will see below, the distributions in Eq. 4 and Eq.

5 [↗](#) can be experimentally verified when appropriate measurements are available. Finally, we define the population average of the cell state dependent mutual information:

$$I_{\text{Cee}} = \int I(\theta) p(\theta) d\theta. \quad (6)$$

In information theory, I_{Cee} is known as the conditional mutual information³⁷ [↗](#) between the input u and the output x conditioned on θ . If cell state variables remain constant and if distribution over cell states is independent of the input distribution, it can be shown that $I_{\text{Cee}} \geq I$ (SI Section 1). It then follows that at least some cells in the population have better sensing abilities compared to the cell state agnostic mutual information (Eq. 1 [↗](#)).

Since I_{Cee} depends on the input distribution $p(u)$, we can find an optimal input distribution $p(u)$ that maximizes I_{Cee} (SI Section 1). Going forward, unless an input distribution is specified, the distributions $p_{\text{CeeMI}}(I)$ and $p_{\text{CeeMI}}(I, \chi)$ are discussed in the context of this optimal input distribution.

Maximum entropy inference can estimate $p_{\text{CeeMI}}(I)$

Estimation of $p_{\text{CeeMI}}(I)$ requires decomposing the experimentally observed response $p(x|u)$ into cell specific output distributions $p(x|u, \theta)$ and the distribution of cell state variables $p(\theta)$ (Eq. 3 [↗](#) and 4 [↗](#)). This problem is difficult to solve given that neither $p(x|u, \theta)$ nor $p(\theta)$ are accessible in typical experiments. However, for many signaling networks, stochastic biochemical models can be constructed to obtain the cell specific output distribution $p(x|u, \theta)$. Here, θ comprise protein concentrations and effective rates of enzymatic reactions and serve as a proxy for cell state variables. Given the experimentally measured cell state averaged response $p(x|u)$ and the model predicted cell specific output distribution $p(x|u, \theta)$, we need a computational method to infer $p(\theta)$ (see Eq. 2 [↗](#)). The problem of inference of parameter heterogeneity is a non-trivial inverse problem³⁷ [↗](#). While there are several proposed methods to solve this problem (reviewed in³⁸ [↗](#)), most cannot efficiently infer $p(\theta)$ for signaling networks with even a moderate ($n \sim 10$) number of parameters. Here, we use our previously developed maximum entropy-based approach to infer $p(\theta)$ ³⁷ [↗](#). This way, we can estimate $p_{\text{CeeMI}}(I)$ using experimentally obtained cell state agnostic response $p(x|u)$ and a stochastic biochemical model $p(x|u, \theta)$ of the underlying signaling network.

An “average cell” can discern much less than a typical cell about the environment

To illustrate the effect of heterogeneity of cell state variables on the cell state agnostic estimate of mutual information (which we call I_{CSA} from now onwards), we consider a simple stochastic biochemical network of a receptor-ligand system. Details of the model and the calculations presented below can be found in SI Section 2. Briefly, extracellular ligand L binds to cell surface receptors R . Steady state levels of the ligand bound receptor B is considered the output. The signaling network is parametrized by several cell state variables θ such as receptor levels, rates of binding/unbinding, etc. For simplicity, we assume that only one variable, steady state receptor level R_0 in the absence of the ligand, varies in the population. Calculations with variability in other parameters are presented in SI Section 2.

In this population, cells’ response $p(B|L, R_0)$ is distributed as a Poisson distribution whose mean is proportional to the cell state variable R_0 (SI Section 2). That is, when all other parameters are fixed, a higher R_0 leads to lower noise (coefficient of variation). To calculate cell state dependent mutual information (Eq. 3 [↗](#)), we assume that $p(L)$ is a gamma distribution. As expected, $I(R_0)$ (Eq. 3 [↗](#)) between the output B and the input L increases monotonically with R_0 (inset in Fig. 2A [↗](#)). Moreover, given that R_0 varies in the population (also assumed to be a gamma distribution), the sensing ability varies in the population as well (Fig. 2A [↗](#)). Notably, the average I_{Cee} of $I(R_0)$

remains relatively robust to variation in. At the same time, the traditional estimate, which is the mutual information between the input and the cell state agnostic population response (response of the “average” cell, *Eq. 1* [↗](#) and *Eq. 2* [↗](#)), decreases as the population heterogeneity in increases. Importantly, is significantly lower than the sensing ability of most cells in the population (**Fig. 2A** [↗](#)). This is because the overlap in the population response distributions is significantly larger than that in single cell response distributions (**Fig. 2B** [↗](#)). This simple example illustrates that the traditional mutual information estimates may severely underestimate cells’ ability to resolve inputs, especially when cell state variables are heterogeneously distributed. Moreover, it is crucial to explicitly account for heterogeneity in cell state variables when estimating fidelity of cellular communication channels.

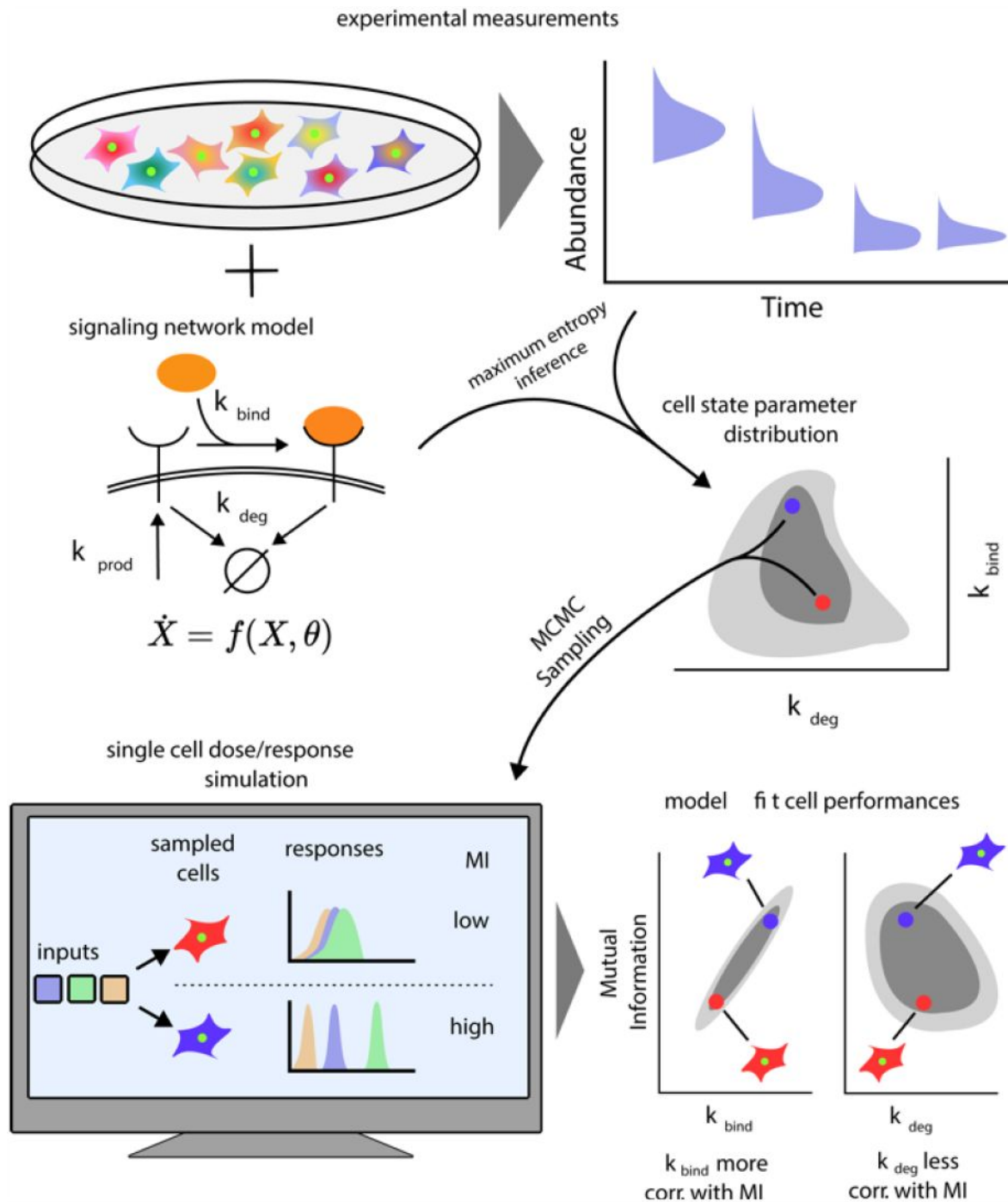
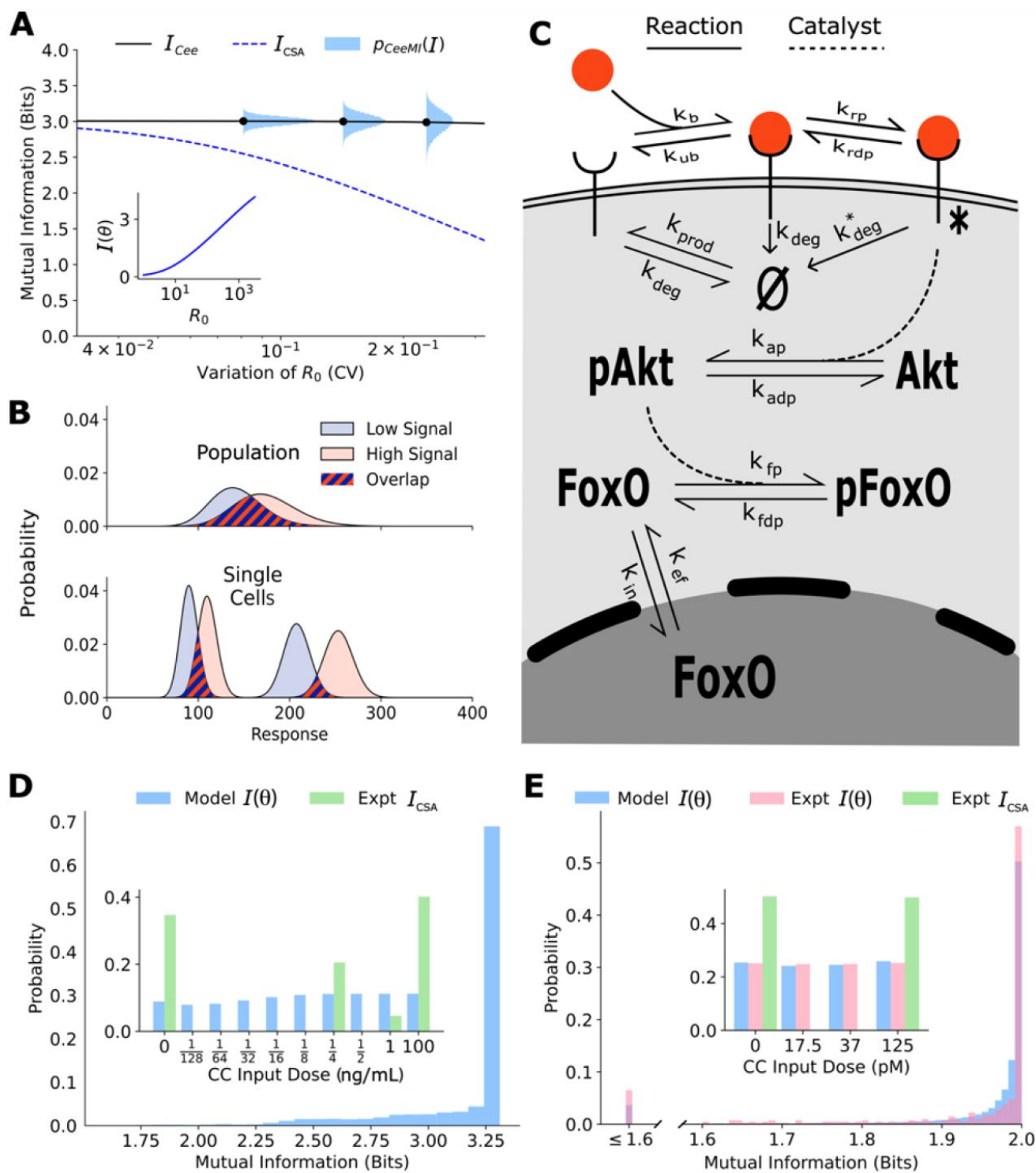


Figure 1.

A schematic of our computational approach.

(top) single cell data across different input conditions and time points are integrated with a stochastic model of a signaling network using a previously developed maximum entropy approach leading to a distribution over signaling network parameters $p(\theta)$ (middle). (bottom) In silico cells are generated using the inferred parameter distribution and cell-state specific mutual information $I(\theta)$ and population distribution of cell performances $p_{\text{CeeMI}}(I)$ is estimated. The model also evaluates the correlation between cells' performance and biochemical parameters.



collected single cell data on the two pathways³⁷,⁴¹,⁴³ and our maximum entropy-based framework³⁷, we estimated the distribution over parameters $p(\theta)$ for the two pathways (SI Section 3). Using these inferred distributions, and the model-predicted cell state specific response distribution $p(x|u, \theta)$, we could compute $p_{\text{CeeMI}}(I)$ for any specified input distribution $p(u)$. We choose the support of the input distribution as the ligand concentrations used in the experimental setup. The estimates shown in **Fig. 2D** and **Fig. 2E** show $p_{\text{CeeMI}}(I)$ corresponding to the input distribution $p(u)$ that maximizes the conditional mutual information I_{Cee} (see Eq. 6). This input distribution is shown in the inset of **Fig. 2D** and **2E**.

Similar to the illustrative example (**Fig. 2A**), there is a wide distribution of single cell sensing fidelities in real populations (**Fig. 2D** and **Fig. 2E**). Moreover, most cells are better sensors compared to the “average cell”, a cell whose response $p(x|u)$ is averaged over cell state variability which was estimated to be ~ 1 bit for both pathways. The cellular signaling fidelity skews towards the upper limit of 2 bits which corresponds to the logarithm of the number of inputs used in the experiment. Indeed, as seen in the insets of **Fig. 2D** and **Fig. 2E**, the input distribution corresponding to the maximum of the cell state agnostic mutual information I_{CSA} is concentrated on the lowest and the highest input for both pathways, indicating that cells may be able to detect only two input levels. In contrast, the input distribution corresponding to the maximum of I_{Cee} is close to uniform, suggesting that individual cells can in fact resolve different ligand levels.

To verify our computational estimate of $p_{\text{CeeMI}}(I)$ for the IGF/FoxO pathway, we reanalyzed previously collected data wherein several cells were stimulated using successive IGF levels⁴¹. The details of our calculations can be found in SI Section 4. Briefly, the cells reached an approximate steady state within 60 minutes of each stimulation and nuclear FoxO levels measured between 60 to 90 minutes were used to approximate an experimental cell state specific response distribution $p(x|u, \theta)$. The distribution $p_{\text{CeeMI}}(I)$ was then obtained by maximizing the average mutual information I_{Cee} (averaged over all cells) with respect to the input distribution. As seen in **Fig. 2D** and **Fig. 2E**, the experimentally evaluated single cell signaling fidelities match closely with our computational estimates. Moreover, as predicted using our computational analysis, individual cells in the population were significantly better at sensing than what is implied by the maximum of I_{CSA} . Indeed, the distribution of steady state FoxO levels were found to be well-resolved at the single cell level as well (**Fig. 2E**). Live cell imaging data for the EGFR pathway was not available and we leave it to future studies to validate our predictions.

Our calculations show that real cell populations comprise cells that have differing sensing fidelities, individual cells are significantly better at sensing their environment than what traditional estimates would indicate, and importantly, the CeeMI approach can accurately estimate the variation in signaling performances using readily collected experimental data and stochastic biochemical models. Notably, the variability in cell states and therefore the heterogeneity in sensing abilities is likely to be stable over time; the same cell's FoxO responses to the same input were found to have significantly less variation compared to the variation within the population (SI Section 4).

CeeMI identifies biochemical parameters that determine cells' ability to sense their environment

Like other phenotypes²¹ and signaling dynamics²⁰,³⁰, we expect that cells' ability to sense their environment depends on their state variables. For example, cells with faster endocytosis rates³⁹ may integrate environmental fluctuations with higher accuracy⁴³. Or cells with higher receptor numbers²¹ will lead to a lowered relative noise in bound ligand concentration.

To systematically identify the variables that differentiate between cells' ability to sense the environment, we quantify the joint distribution $p_{\text{CeeMI}}(I, \chi)$ of single cell signaling fidelity and biochemical state variables that take part in the signaling network. To test whether we can

identify variables that are predictive of cellular fidelity, we estimated the joint distribution $p_{\text{CeeMI}}(I, \chi)$ for two variables that were experimentally accessible, response range of nuclear FoxO (**Fig. 3** [left](#), see inset) and total nuclear FoxO levels prior to IGF stimulation (**Fig. 3** [right](#), see inset). In both figures, the shaded regions show computational estimates of the joint probability distributions, and the red circles represent real cells. The green and the cyan trend lines represent computational and experimental binned averages.

One may expect that higher total nuclear FoxO levels could result in lower noise (coefficient of variation) and therefore better sensing abilities. However, we find that total nuclear FoxO levels only weakly correlate with cell state dependent mutual information (correlation coefficient $r = 0.16$ for computational estimates and $r = 0.04$ for experimental data). In contrast, cell state dependent mutual information depended strongly on the dynamic range of the response (correlation coefficient $r = 0.53$ for computational estimates and $r = 0.29$ for experimental data). Importantly, the model captured the observation that cells with a small response range had a variable sensing abilities while cells with a large response range all performed well in resolving extracellular IGF levels. Surprisingly, the total nuclear FoxO levels only weakly correlated with the cell specific mutual information. In SI Section 5, we show the predicted joint distributions $p_{\text{CeeMI}}(I, \chi)$ for several other biochemical variables that can potentially govern single cells' response to extracellular IGF stimuli. This way, CeeMI can be used to systematically identify cell state variables that differentiate between good sensors and bad sensors of the environmental stimuli.

Time scale of stochastic dynamics of cell states dictates the divergence between state-specific and state-agnostic sensing abilities

A key limitation of the presented analysis is the assumption that cell state variables remain approximately constant over the time scale of typical environmental fluctuations. While many cell state variables change slowly, remaining roughly constant over the life spans of cells and beyond²⁵[, 26](#), state changes may occur within the lifespan of a cell as well⁴⁵. These dynamical changes can be accommodated easily in our calculations. Here, instead of fixing the cell state variables θ , we can treat them as initial conditions and propagate them stochastically with their own dynamics. In the limit of very fast dynamics where individual cells rapidly transition cell states, we expect that the individuality of cells in a population vanishes and consequently cell state specific mutual information for each cell will agree with traditional cell state agnostic estimates of the channel capacity. In contrast, if the cell state dynamics are slow, our framework highlights differences between cells in the population.

To elucidate the role of cell state dynamics, we built a simple model of ligand-receptor system (SI Section 6, **Fig. 4** [left](#)). Briefly, the model included production and degradation of the receptor mRNA, translation of the receptor protein from the mRNA, and ligand binding to the receptor (**Fig. 4A** [left](#)). The number of ligand-free receptors and the number mRNA molecules were together considered the state variables. We tuned the mRNA dynamics by keeping the average mRNA number constant while simultaneously changing mRNA production and degradation rates. The time scale of mRNA dynamics is denoted by τ .

When the mRNA dynamics are slower than translation, ligand binding, and receptor degradation, individual cells are effectively frozen around a particular receptor number (**Fig. 4B** [left](#)). The distribution of the number of ligand-bound receptors after a short-term ligand exposure reflects this frozen state. As a result, the cell-state conditioned mutual information depends strongly the cell state, with higher mutual information for cells with higher receptor numbers. In contrast, as the mRNA dynamics become faster, cells quickly lose the memory of cell state conditioning. As a result, the distribution of the number of ligand-bound receptors after a short-term ligand exposure reflects an ergodic averaging over the underlying mRNA dynamics. This averaging effectively washes out differences between cells (**Fig. 4C** [left](#)).

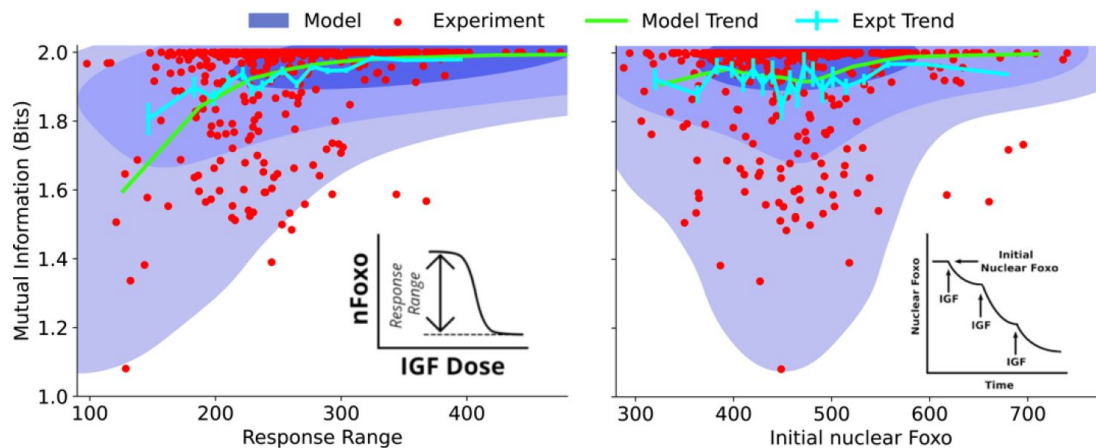


Figure 3.

Dependence on cell state dependent mutual information on biochemical parameters.

(left) The joint distribution $p_{\text{CeeMI}}(I, \chi)$ of cell state specific mutual information and biochemical parameter χ chosen to be the single cell response range of nuclear FoxO levels (x-axis, see inset for a cartoon). The shaded blue regions are model predictions, and the green line is the model average. The darker shades represent higher probabilities. The red dots represent experimental cells. The cyan line represents experimental averages. (right) same as (left) with biochemical parameter χ chosen to be steady state nuclear Foxo levels in the absence of stimulation. The contours represent 1% to 10%, 10% to 50%, and 50% to 100% of the total probability mass (from faint to dark shading).

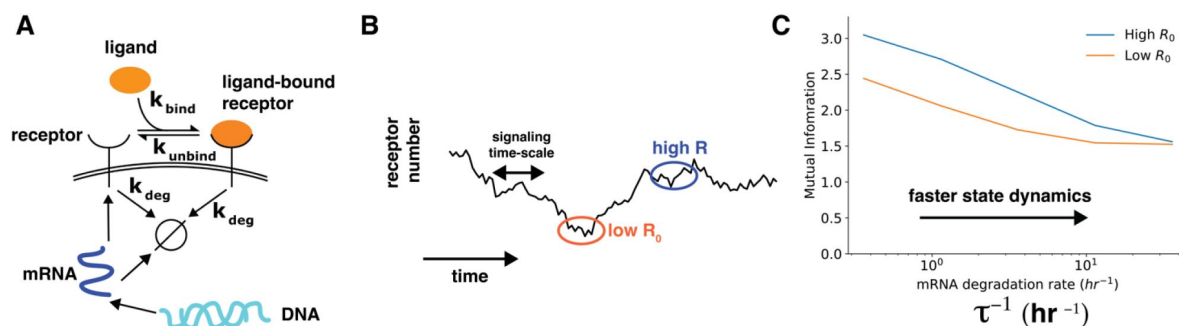


Figure 4.

Cell state dynamics governs cell state conditioned mutual information.

A. In a simple stochastic model, receptor mRNA is produced at a constant rate from the DNA and the translated into ligand-free receptors. The number of ligand-bound receptors after a short exposure to ligands is considered the output. B. A schematic showing dynamics of receptor numbers when mRNA dynamics are slower compared to signaling time scales. C. Conditioning on receptor numbers leads to differing abilities in sensing the environment when the time scale of mRNA dynamics is slow. In contrast, when the mRNA dynamics are fast (large τ^{-1}), conditioning on cell state variables does not lead to difference in sensing abilities.

This example shows that when cells states change at time scales slower than signaling dynamics, cells in a population can be identified by their cell state and their ability to sense the environment can differ from one another.

Discussion

Cell populations are characterized by heterogeneity in cell state variables^{18, 19} that is responsible for important phenotypic differences and selective advantage for example in sensitivity to drugs^{25, 26, 28}, response to chemotactic signals⁴⁶, and following proliferation cues²⁹. Therefore, it is reasonable to expect that cells' ability to sense their environment depends on their state and is therefore variable across cells in a population. To quantify this heterogeneity, here, we developed a novel information theoretic perspective that allowed us to quantify the distribution $p_{\text{CeeMI}}(I)$ of single cell sensing abilities using easily measurable single cell data and stochastic models of signaling networks. We also quantified the joint distribution $p_{\text{CeeMI}}(I, \chi)$ of cell specific sensing ability and biochemical cell state variable. Importantly, using two growth factor pathways, we showed that individual cells in real cell populations were much better at sensing their environment compared to what is implied by the traditional estimate of channel capacities of signaling networks. Typical single cell data are time stamped and do not give information about dynamics in the single cell. Moreover, these data do not provide information about dependence of cell-to-cell variability on cell state variables. Therefore, computational modeling of cellular trajectories from single cell data and stochastic models is essential in deciphering mechanistic origins of heterogeneity in cellular phenotypes.

The approach presented here will be useful in identifying bottlenecks in signal transduction. Many cellular phenotypes such as chemotaxis^{7, 8} and cell proliferation⁴⁷ exhibit a weak correlation between cellular outputs (e.g., directional alignment with chemical gradients) with the input (e.g., gradient strength), resulting in a low channel capacity even for individual cells. In such cases, it is important to understand where exactly along the information transduction pathway is the information about the gradient is lost. If traditional calculations are pursued, for example, for movement of mammalian cells under growth factor gradients^{7, 31}, one may conclude that the information loss likely happens right at the receptor level (**Fig. 2D**). In contrast, CeeMI will allow us to disentangle the effect of cell state heterogeneity and noisy cellular response to precisely pinpoint intracellular signaling nodes that are responsible for signal corruption.

How do we contrast our results with previous low estimates of channel capacities? There is extensive population heterogeneity in cell state variables and this heterogeneity often is stable over time. Nonetheless, this heterogeneity arises from stochasticity in underlying processes including dynamics of epigenetic transitions and gene expression. When one specifies state variables¹⁹ θ , one effectively conditions on the stochastic trajectory of the underlying dynamics to a specific narrow window (**Fig. 4**). This extra conditioning narrows the cells' response distributions, revealing the heterogeneous nature of cell populations¹⁹.

In summary, we showed that like other phenotypes, the ability to sense the environment is itself heterogeneously distributed in a cell population. Moreover, we also showed that when conditioned on cell state variables, mammalian cells appear to be significantly better at sensing their environment than what traditional mutual information calculation suggests. Finally, we showed that we could identify cell state variables that made some cells better sensors compared to others. We believe that CeeMI will be an important framework in quantifying fidelity of input/output relationships in heterogeneous cell populations.

Acknowledgements

AG, HA, and PD are supported by NIGMS grant R35GM142547. The authors would like to thank Andre Levchenko for useful discussions.

References

1. Levchenko A, Nemenman I. (2014) **Cellular noise and information transmission** *Curr Opin Biotechnol* **28**:156–64
2. Rhee A, Cheong R, Levchenko A. (2012) **The application of information theory to biochemical signaling systems** *Phys Biol* **9**
3. en Wolde PR, Becker NB, Ouldridge TE, Mugler A. (2016) **Fundamental Limits to Cellular Sensing** *J Stat Phys* **162**:1395–424
4. Cheong R, Rhee A, Wang CJ, Nemenman I, Levchenko A. (2011) **Information transduction capacity of noisy biochemical signaling networks** *Science* **334**:354–8
5. Selimkhanov J, Taylor B, Yao J, Pilko A, Albeck J, Hoffmann A, Tsimring L, Wollman R. (2014) **Accurate information transmission through dynamic biochemical signaling networks** *Science* **346**:1370–3
6. Suderman R, Bachman JA, Smith A, Sorger PK, Deeds EJ (2017) **Fundamental trade-offs between information flow in single cells and cellular populations** *P Natl Acad Sci USA* **114**:5755–60
7. Varennes J, Moon HR, Saha S, Mugler A, Han B. (2019) **Physical constraints on accuracy and persistence during breast cancer cell chemotaxis** *Plos Comput Biol* **15**
8. Mattingly HH, Kamino K, Machta BB, Emonet T. (2021) **Escherichia coli chemotaxis is information limited** *Nat Phys* **17**
9. Cover TM, Thomas JA, Hoboken N.J. (2006) **Elements of information theory**
10. Wu M, Singh AK (2012) **Single-cell protein analysis** *Curr Opin Biotechnol* **23**:83–8
11. Slavov N. (2021) **Single-cell protein analysis by mass spectrometry** *Curr Opin Chem Biol* **60**:1–9
12. Lemon WC, McDole K. (2020) **Live-cell imaging in the era of too many microscopes** *Curr Opin Cell Biol* **66**:34–42
13. Hansen AS, O’Shea EK (2015) **Limits on information transduction through amplitude and frequency regulation of transcription factor activity** *Elife* **4**
14. Billing U, Jetka T, Nortmann L, Wundrack N, Komorowski M, Waldherr S, Schaper F, Dittrich A. (2019) **Robustness and Information Transfer within IL-6-induced JAK/STAT Signalling** *Commun Biol* **2**
15. Garner KL, Perrett RM, Voliotis M, Bowsher C, Pope GR, Pham T, Caunt CJ, Tsaneva-Atanasova K, McArdle CA (2016) **Information Transfer in Gonadotropin-releasing Hormone (GnRH) Signaling: EXTRACELLULAR SIGNAL-REGULATED KINASE (ERK)-MEDIATED FEEDBACK LOOPS CONTROL HORMONE SENSING** *J Biol Chem* **291**:2246–59

16. Frick CL, Yarka C, Nunns H, Goentoro L. (2017) **Sensing relative signal in the Tgf-beta/Smad pathway** *Proc Natl Acad Sci U S A* **114**:E2975–E82
17. Lee JB, Caywood LM, Lo JY, Levering N, Keung AJ (2021) **Mapping the dynamic transfer functions of eukaryotic gene regulation** *Cell Syst* **12**:1079–93
18. Trapnell C. (2015) **Defining cell types and states with single-cell genomics** *Genome Res* **25**:1491–8
19. Wollman R. (2018) **Robustness, Accuracy, and Cell State Heterogeneity in Biological Systems** *Curr Opin Syst Biol* **8**:46–50
20. Meyer R *et al.* (2012) **Heterogeneous kinetics of AKT signaling in individual cells are accounted for by variable protein concentration** *Front Physiol* **3**
21. Frankel NW, Pontius W, Dufour YS, Long J, Hernandez-Nunez L, Emonet T. (2014) **Adaptability of non-genetic diversity in bacterial chemotaxis** *Elife* **3**
22. Chen KH, Boettiger AN, Moffitt JR, Wang S, Zhuang X. (2015) **RNA imaging. Spatially resolved, highly multiplexed RNA profiling in single cells** *Science* **348**
23. Nagano T, Lubling Y, Stevens TJ, Schoenfelder S, Yaffe E, Dean W, Laue ED, Tanay A, Fraser P. (2013) **Single-cell Hi-C reveals cell-to-cell variability in chromosome structure** *Nature* **502**:59–64
24. Buenrostro JD, Wu B, Litzenburger UM, Ruff D, Gonzales ML, Snyder MP, Chang HY, Greenleaf WJ (2015) **Single-cell chromatin accessibility reveals principles of regulatory variation** *Nature* **523**:486–90
25. Flusberg DA, Roux J, Spencer SL, Sorger PK (2013) **Cells surviving fractional killing by TRAIL exhibit transient but sustainable resistance and inflammatory phenotypes** *Mol Biol Cell* **24**:2186–200
26. Spencer SL, Gaudet S, Albeck JG, Burke JM, Sorger PK (2009) **Non-genetic origins of cell-to-cell variability in TRAIL-induced apoptosis** *Nature* **459**:428–32
27. Pleska M, Jordan D, Frentz Z, Xue B, Leibler S. (2021) **Nongenetic individuality, changeability, and inheritance in bacterial behavior** *Proc Natl Acad Sci U S A* **118**
28. Emert BL, Cote CJ, Torre EA, Dardani IP, Jiang CL, Jain N, Shaffer SM, Raj A. (2021) **Variability within rare cell states enables multiple paths toward drug resistance** *Nat Biotechnol* **39**:865–76
29. Matson JP, Cook JG (2017) **Cell cycle proliferation decisions: the impact of single cell analyses** *FEBS J* **284**:362–75
30. Norris D *et al.* (2021) **Signaling Heterogeneity is Defined by Pathway Architecture and Intercellular Variability in Protein Expression** *Iscience* **24**
31. Moon HR, Saha S, Mugler A, Han B. (2021) **Signal processing capacity of the cellular sensory machinery regulates the accuracy of chemotaxis under complex cues** *Iscience* **24**
32. Waite AJ, Frankel NW, Dufour YS, Johnston JF, Long J, Emonet T. (2016) **Non-genetic diversity modulates population performance** *Mol Syst Biol* **12**

33. Clark JA, Campbell SG (2019) **Diverse relaxation rates exist among rat cardiomyocytes isolated from a single myocardial region** *J Physiol* **597**:711–22
34. Yao J, Pilko A, Wollman R. (2016) **Distinct cellular states determine calcium signaling response** *Mol Syst Biol* **12**
35. Lewis Cantley TH (2013) **Richard Sever, Jeremy Thorner Signal Transduction: Principles, Pathways, and Processes**
36. Feinberg AP, Levchenko A. (2023) **Epigenetics as a mediator of plasticity in cancer** *Science* **379** <https://doi.org/10.1126/science.aaw3835>
37. Dixit PD, Lyashenko E, Niepel M, Vitkup D. (2020) **Maximum Entropy Framework for Predictive Inference of Cell Population Heterogeneity and Responses in Signaling Networks** *Cell Syst* **10**:204–12
38. Loos C H J. (2019) **Mathematical modeling of variability in intracellular signaling** *Current Opinion in Systems Biology* **16**:17–24
39. Kallenberger SM, Unger AL, Legewie S, Lymeropoulos K, Klingmuller U, Eils R, Hertzen DP (2017) **Correlated receptor transport processes buffer single-cell heterogeneity** *Plos Comput Biol* **13**
40. Chung JC, Sciaky N, Gross DJ (1997) **Heterogeneity of epidermal growth factor binding kinetics on individual cells** *Biophys J* **73**:1089–102
41. Gross SM, Dane MA, Bucher E, Heiser LM (2019) **Individual Cells Can Resolve Variations in Stimulus Intensity along the IGF-PI3K-AKT Signaling Axis** *Cell Syst* **9**:580–8
42. Wimmer RJ, Liu Y, Schachter TN, Stonko DP, Peercy BE, Schneider MF (2014) **Mathematical modeling reveals modulation of both nuclear influx and efflux of Foxo1 by the IGF-I/PI3K/Akt pathway in skeletal muscle fibers** *Am J Physiol Cell Physiol* **306**:C570–84
43. Lyashenko E, Niepel M, Dixit PD, Lim SK, Sorger PK, Vitkup D. (2020) **Receptor-based mechanism of relative sensing and cell memory in mammalian signaling networks** *Elife* **9**
44. Fortian A, Sorkin A. (2014) **Live-cell fluorescence imaging reveals high stoichiometry of Grb2 binding to the EGF receptor sustained during endocytosis** *J Cell Sci* **127**:432–44
45. MacLean AL, Hong T, Nie Q. (2018) **Exploring intermediate cell states through the lens of single cells** *Curr Opin Syst Biol* **9**:32–41
46. Mattingly HH, Emonet T. (2022) **Collective behavior and nongenetic inheritance allow bacterial populations to adapt to changing environments** *Proc Natl Acad Sci U S A* **119**
47. Gross SM, Rotwein P. (2016) **Unraveling Growth Factor Signaling and Cell Cycle Progression in Individual Fibroblasts** *J Biol Chem* **291**:14628–38

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Reviewer #2 (Public Review):

In this paper the authors present an existing information theoretic framework to assess the ability of single cells to encode external signals sensed through membrane receptors. The main point is to distinguish actual noise in the signaling pathway from cell-cell variability, which could be due to differences in their phenotypic state, and to formalize this difference using information theory. After correcting for this cellular variability, the authors find that cells may encode more information than one would estimate from ignoring it, which is expected. The authors show this using simple models of different complexities, and also by analyzing an imaging dataset of the IGF/FoxO pathway.

I am only partially satisfied by the authors response. To me, the main question that was unanswered, while being at the core of the claim of the paper, was the magnitude of within-cell variability across repetitions of the stimulus.

This can only be done on the IGF/FoxO system because, as the authors acknowledge, the EGF/EGFR system does not have any data to support any claim about single-cell information that's not heavily informed by models, which assume by construction that this variability is small, naturally leading the desired conclusion.

The authors now measure within-cell, across-repetition variability (Δ_0) for IGF/FoxO, but:

- they compare it to cell-to-cell variability, finding that it's smaller. That's good and that supports the main claim of the paper that single cells are more precise than a mean cell. However they don't show it in the paper, but only in the response.
- they also don't compare it to within-cell, within-stimulation variability across time. But this latter variability is what they (wrongly) used to estimate information, and still do in this

revision. However I think this is approximated by the blue "simulation" violin plot in Reviewer Figure 2. The true variability is clearly larger than previously assumed. So it's strange that they conclude that "our estimates of cell-to-cell variability signaling fidelity are stable and reliable."

- they don't use this Δ_0 variability to revise their estimate of the information accordingly.
- since variability is small compared to the differences between distinct stimulations, of which there are only 4, all information quantities they get are around 2 bits, which is not approaching the information capacity but merely a statement that the number of tested doses is small.

I strongly recommend that the authors actually report the figure they provided as Reviewer Figure 2 in the manuscript. In addition, they should not claim that the within-cell variability (captured by the variability across distinct presentations of the stimulus) is well captured by their initial estimate (based on the variance within a single presentation of the stimulus).

- <https://doi.org/10.7554/eLife.87747.2.sa1>

Reviewer #3 (Public Review):

Goetz, Akl and Dixit investigated the heterogeneity in the fidelity of sensing the environment by individual cells in a population using computational modeling and analysis of experimental data for two important and well-studied mammalian signaling pathways: (insulin-like growth factor) IGF/FoxO and (epidermal growth factor) EFG/EGFR mammalian pathways. They quantified this heterogeneity using the conditional mutual information between the input (eg. level of IGF) and output (eg. level of FoxO in the nucleus), conditioned on the "state" variables which characterize the signaling pathway (such as abundances of key proteins, reaction rates, etc.) First, using a toy stochastic model of a receptor-ligand system - which constitutes the first step of both signaling pathways - they constructed the population average of the mutual information conditioned on the number of receptors and maximized over the input distribution and showed that it is always greater than or equal to the usual or "cell state agnostic" channel capacity. They constructed the probability distribution of cell state dependent mutual information for the two pathways, demonstrating agreement with experimental data in the case of the IGF/FoxO pathway using previously published data. Finally, for the IGF/FoxO pathway, they found the joint distribution of the cell state dependent mutual information and two experimentally accessible state variables: the response range of FoxO and total nuclear FoxO level prior to IGF stimulation. In both cases, the data approximately follow the contour lines of the joint distribution. Interestingly, high nuclear FoxO levels, and therefore lower associated noise in the number of output readout molecules, is not correlated with higher cell state dependent mutual information, as one might expect. This paper contributes to the vibrant body of work on information theoretic characterization of biochemical signaling pathways, using the distribution of cell state dependent mutual information as a metric to highlight the importance of heterogeneity in cell populations. The authors suggest that this metric can be used to infer "bottlenecks" in information transfer in signaling networks, where certain cell state variables have a lower joint distribution with the cell state dependent mutual information.

The utility of a metric based on the conditional mutual information to quantify fidelity of sensing and its heterogeneity (distribution) in a cell population is supported in the comparison with data. Some aspects of the analysis and claims in the main body of the paper and SI need to be clarified and extended.

Remaining Comments:

- I think Review Figure 2 which is currently in the SI would improve the main body of the paper if moved there. In that case, the discussion of this figure in the main text would have to

address more than it currently does, namely "the same cell's FoxO responses to the same input were found to have significantly less variation compared to the variation within the population".

- <https://doi.org/10.7554/eLife.87747.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

The manuscript by Goetz et al. takes a new perspective on sensory information processing in cells. In contrast to previous studies, which have used population data to build a response distribution and which estimate sensory information at about 1 bit, this work defines sensory information at the single cell level. To do so, the authors take two approaches. First, they estimate single cells' response distributions to various input levels from time-series data directly. Second, they infer these single-cell response distributions from the population data by assuming a biochemical model and extracting the cells' parameters with a maximum-entropy approach. In either case, they find, for two experimental examples, that single-cell sensory information is much higher than 1 bit, and that the reduction to 1 bit at the population level is due to the fact that cells' response functions are so different from each other. Finally, the authors identify examples of measurable cell properties that do or do not correlate with single-cell sensory information.

The work brings an important and distinct new insight to a research direction that generated strong interest about a decade ago: measuring sensory information in cells and understanding why it is so low. The manuscript is clear, the results are compelling, and the conclusions are well supported by the findings. Several contributions should be of interest to the quantitative biology community (e.g., the demonstration that single cells' sensory information is considerably larger than previously implied, and the approach of inferring single-cell data from population data with the help of a model and a maximum-entropy assumption).

We thank the reviewer for the excellent summary of our research.

Reviewer #2 (Public Review):

In this paper the authors present an existing information theoretic framework to assess the ability of single cells to encode external signals sensed through membrane receptors.

The main point is to distinguish actual noise in the signaling pathway from cell-cell variability, which could be due to differences in their phenotypic state, and to formalize this difference using information theory.

After correcting for this cellular variability, the authors find that cells may encode more information than one would estimate from ignoring it, which is expected. The authors show this using simple models of different complexities, and also by analyzing an imaging dataset of the IGF/FoxO pathway.

The implications of the work are limited because the analysed data is not rich enough to draw clear conclusions. Specifically,

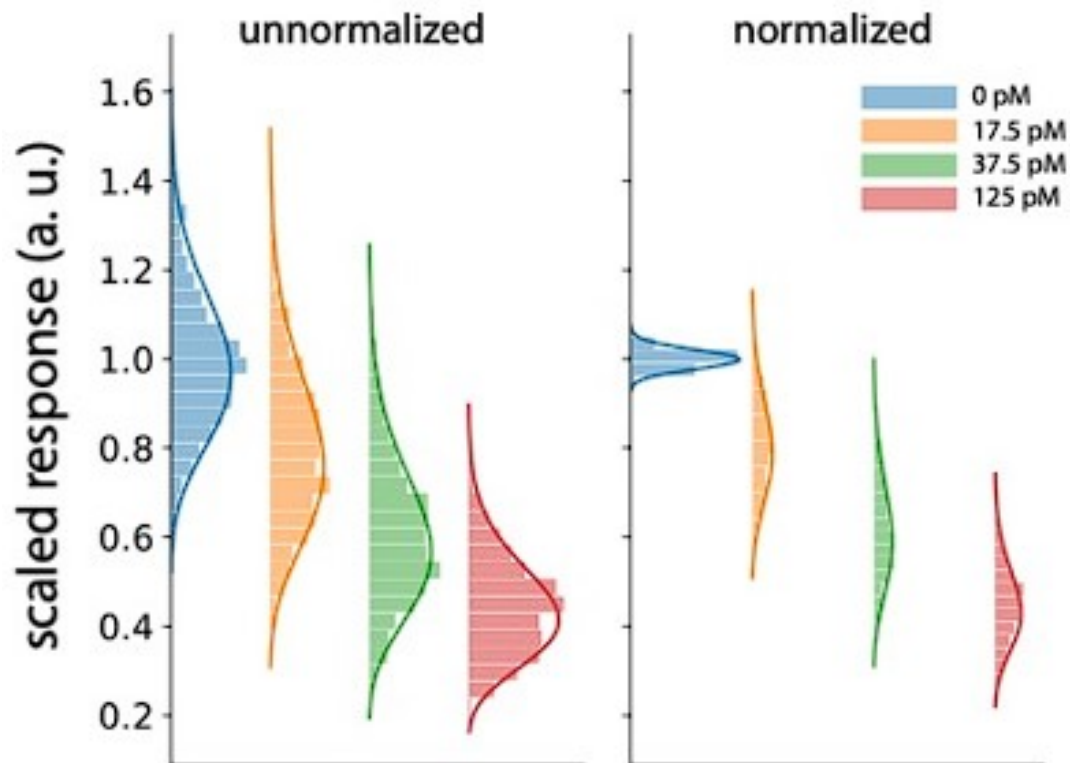
- *the authors do not distinguish what could be methodological noise inherent to microscopy techniques (segmentation etc), and actual intrinsic cell state. It's not clear that cell-cell variability in the analyzed dataset is not just a constant offset or normalization factor. Other authors (e.g. Gregor et al Cell 130, 153-164) have re-centered and re-normalized their data before further analysis, which is more or less equivalent to the idea of the conditional information in the sense that it aims to correct for this experimental noise.*

We thank the reviewer for the comment. However, we do not believe our analysis is a consequence of normalization artifacts. Prior to modeling the single cell data, we removed well-dependent background fluorescence. This should take care of technical variation related to overall offsets in the data. We agree with the reviewer that background subtraction may not fully account for technical variability. For example, some of the cell-to-cell variability may potentially be ascribed to issues such as incorrect segmentation. Unfortunately, however, attempting to remove this technical variability through cell-specific normalization as suggested by the reviewer¹ will diminish to a very large extent the true biological effects related to extensivity (cell size, total protein abundance). We note that these effects are a direct function of cell state-variables (see for example Cohen-Saidon et al.² who use cell-state specific normalization to improve signaling fidelity). Therefore, an increase in mutual information after normalization does not only reflect removal of technical noise but also accounts for effect of cell state variables.

Nonetheless, as the reviewer suggested, we performed a cell-specific normalization wherein the mean nuclear FoxO levels in each cell (in the absence of IGF) were normalized to one. Then, for each ligand concentration, we collated FoxO response across all cells and computed the channel capacity corresponding to cell-state agnostic mutual information ICSA. As expected, ICSA increases from ~0.9 bits to ~1.3 bits when cell-specific normalization was performed (Author response image 1). However, this value is significantly lower than the average ~1.95 of cell-state specific mutual information (IC_{cell}). Finally, we note that the cell specific normalization does not change the calculations of channel capacity at the single cell level as these calculations do not depend on linear transformations of the data (centering and normalization). Therefore, we do not think that our analysis of experimental data suffers from artifacts related to microscopy.

Author response image 1.

Author response image 1. Left: nuclear FoxO response averaged over all cells in the population across different ligand concentration. Right: nuclear FoxO response was first normalized at the single cell level and then averaged over all cells in the population across different ligand concentrations.



- *in the experiment, each condition is shown only once and sequentially. This means that the reproducibility of the response upon repeated exposures in a single cell was not tested, casting doubt on the estimate of the response fidelity (estimated as the variance over time in a single response).*

The reviewer raises an excellent question about persistence of cell states. To verify that cell states are indeed conserved at the time scale of the experiment, we reanalyzed data generated by Gross et al.³ wherein cells were perturbed with IGF (37.5 pM), followed by a washout which allowed the cells to reach pre-stimulation nuclear FoxO levels, followed by a re-perturbation with the same amount of IGF. Nuclear FoxO response was measured at the single cell level after 90 minutes with IGF exposure both these times. Since the response x to the same input u was measured twice in the same cell (x_1 and x_2), we could evaluate the intrinsic variability in response at the single cell level. We then compared this intrinsic variability to the extrinsic cell-state dependent variability in the population.

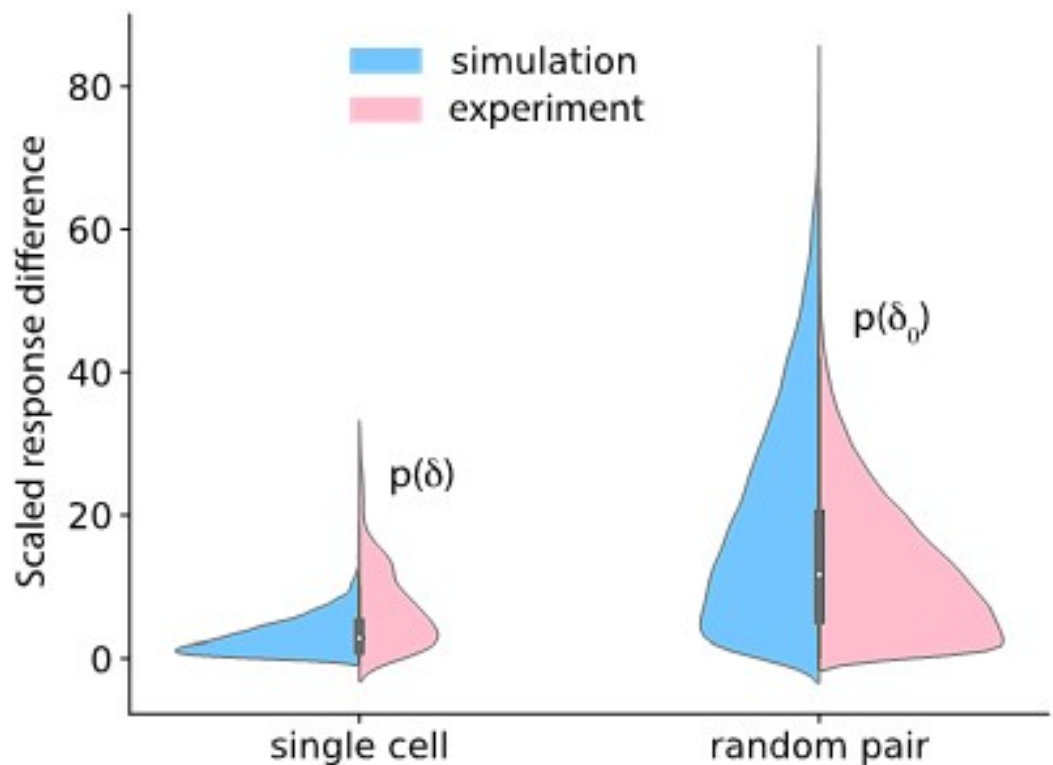
To do so, we computed for each cell $\delta = x_1 - x_2$ the difference between the two responses. reviewer Figure 2 show the histogram $p(\delta)$ as computed from the data (pink) and the same computed from the model that was trained on the single cell data (blue). We also computed $p(\delta_0)$ which represented the difference between responses of two different cells both from the data and from the model.

As we see in Author response image 2, the distribution $p(\delta)$ is significantly narrower than $p(\delta_0)$ suggesting that intracellular variability is significantly smaller than across-population variability and that cells' response to the same stimuli are quite conserved, especially when compared to responses in randomly picked pairs of cells. This shows that cell states and the corresponding response to extracellular perturbations are conserved, at least at the time scale of the experiment. Therefore, our estimates of cell-to-cell variability signaling fidelity

are stable and reliable. We have now incorporated this discussion in the manuscript (lines 275-281).

Author response image 2.

Author response image 2. Left: Cells were treated with 37.5 pM of IGF for 90 minutes, washed out for 120 minutes and again treated with 37.5 pM of IGF. Nuclear FoxO was measured during the treatment and the washout. The distributions on the left show the difference in FoxO levels in single cells after the two 90 minutes IGF stimulations (pink: data, blue: model). Right: Distribution of difference in FoxO levels in two randomly picked cells after 90 minutes of exposure to 37.5 pM IGF.



- another dataset on the EGF/EGFR pathway is analyzed, but no conclusion can be drawn from it because single-cell information cannot be directly estimated from it. The authors instead use a maximum-entropy Ansatz, which cannot be validated for lack of data.

We thank the reviewer for this comment. We agree with the reviewer that we have not verified our predictions for the EGF/EGFR pathway. That study was meant to show the potential generality of our analysis. We look forward to validating our predictions for the EGF/EGFR pathway in future studies.

Reviewer #3 (Public Review):

Goetz, Akl and Dixit investigated the heterogeneity in the fidelity of sensing the environment by individual cells in a population using computational modeling and analysis of experimental data for two important and well-studied mammalian signaling pathways: (insulin-like growth factor) IGF/FoxO and (epidermal growth factor) EGF/EGFR mammalian pathways. They quantified this heterogeneity using the conditional mutual

information between the input (eg. level of IGF) and output (eg. level of FoxO in the nucleus), conditioned on the "state" variables which characterize the signaling pathway (such as abundances of key proteins, reaction rates, etc.) First, using a toy stochastic model of a receptor-ligand system - which constitutes the first step of both signaling pathways - they constructed the population average of the mutual information conditioned on the number of receptors and maximized over the input distribution and showed that it is always greater than or equal to the usual or "cell state agnostic" channel capacity. They constructed the probability distribution of cell state dependent mutual information for the two pathways, demonstrating agreement with experimental data in the case of the IGF/FoxO pathway using previously published data. Finally, for the IGF/FoxO pathway, they found the joint distribution of the cell state dependent mutual information and two experimentally accessible state variables: the response range of FoxO and total nuclear FoxO level prior to IGF stimulation. In both cases, the data approximately follow the contour lines of the joint distribution. Interestingly, high nuclear FoxO levels, and therefore lower associated noise in the number of output readout molecules, is not correlated with higher cell state dependent mutual information, as one might expect. This paper contributes to the vibrant body of work on information theoretic characterization of biochemical signaling pathways, using the distribution of cell state dependent mutual information as a metric to highlight the importance of heterogeneity in cell populations. The authors suggest that this metric can be used to infer "bottlenecks" in information transfer in signaling networks, where certain cell state variables have a lower joint distribution with the cell state dependent mutual information.

The utility of a metric based on the conditional mutual information to quantify fidelity of sensing and its heterogeneity (distribution) in a cell population is supported in the comparison with data. Some aspects of the analysis and claims in the main body of the paper and SI need to be clarified and extended.

1. The authors use their previously published (Ref. 32) maximum-entropy based method to extract the probability distribution of cell state variables, which is needed to construct their main result, namely $p_{\text{CeeMI}}(I)$. The salient features of their method, and how it compares with other similar methods of parameter inference should be summarized in the section with this title. In SI 3.3, the Lagrangian, L , and R_m should be defined.

We thank the reviewer for the comment and apologize for the omission. We have now rewritten the manuscript to include references to previous reviews of works that infer probability distributions⁴ of cell state variables (lines 156-168). Notably, as we argued in our previous work⁵, no current method can efficiently estimate the joint distribution over parameters that is consistent with measured single cell data and models of signaling networks. Therefore, we could not use multiple approaches to infer parameter distributions. We have now expanded our discussion of the method in the supplementary information sections.

1. Throughout the text, the authors refer to "low" and "high" values of the channel capacity. For example, a value of 1-1.5 bits is claimed to be "low". The authors need to clarify the context in which this value is low: In some physically realistic cases, the signaling network may need to simply distinguish between the present or absence of a ligand, in which case this value would not be low.

We agree with the reviewer that small values of channel capacities might be sufficient for cells to carry out some tasks, in which case a low channel capacity does not necessarily indicate a network not performing its task. Indeed, how much information is needed for a specific task is a related but distinct question from how much information is provided though

a signaling network. Both questions are essential to understand a cell's signaling behavior, with the former being far less easy to answer in a way which is generalizable. In contrast, the latter can be quantitatively answered using the analysis presented in our manuscript.

1. Related to (2), the authors should comment on why in Fig. 3A, $I_{Cee}=3$. Importantly, where does the fact that the network is able to distinguish between 23 ligand levels come from? Is this related to the choice (and binning) of the input ligand distribution (described in the SI)?

We thank the reviewer for the comment. The network can distinguish between all inputs used in the in silico experiment precisely because the noise at the cellular level is small enough that there is negligible overlap between single cell response distributions. Indeed, the mutual information will not increase with the number of equally spaced inputs in a sub-linear manner, especially when the input number is very high.

1. The authors should justify the choice of the gamma distribution in a number of cases (eg. distribution of ligand, distribution cell state parameters, such as number of receptors, receptor degradation rate, etc.).

We thank the reviewer for the comment. We note that previous works in protein abundances and gene expression levels (e.g. see6) have reported distributions with positive skews that can be fit well with gamma distributions or log-normal distributions. Moreover, many stochastic models of protein abundance levels and signaling networks are also known to result in abundances that are distributed according to a negative binomial distribution, the discrete counterpart of gamma distribution. Therefore, we chose Gamma distributions in our study. We have now clarified this point in the Supplementary Information. At the same time, gamma distribution only serves as a regularization for the finite data and in principle, our analysis and conclusion do not depend on choice of gamma distribution for abundances of proteins, ligands, and cell parameters.

1. Referring to SI Section 2, it is stated that the probability of the response (receptor binding occupancy) conditioned on the input ligand concentration and number of receptors is a Poisson distribution. Indeed this is nicely demonstrated in Fig. S2. Therefore it is the coefficient of variation (std/mean) that decreases with increasing R_0 , not the noise (which is strictly the standard deviation) as stated in the paper.

We thank the reviewer of the comment. We have now corrected our text.

1. In addition to explicitly stating what the input (IGF level) and the output (nuclear GFP-tagged FoxO level) are, it would be helpful if it is also stated what is the vector of state variables, θ , corresponding to the schematic diagram in Fig. 2C.

We thank the reviewer of the comment. We have now corrected our text in the supplementary material as well as the main text (Figure 2 caption).

1. Related to Fig. 2C, the statement in the caption: "Phosphorylated Akt leads to phosphorylation of FoxO which effectively shuttles it out of the nucleus." needs clarification: From the figure, it appears that pFoxO does not cross the nuclear membrane, in which case it would be less confusing to say that phosphorylation prevents reentry of FoxO into the nucleus.

We thank the reviewer of the comment. We have now corrected our text (Figure 2 caption).

1. The explanations for Fig. 2D, E and insets are sparse and therefore not clear. The authors should expand on what is meant by model and experimental $I(\theta)$. What is CC input dose? Also in Fig. 2E, the overlap between the blue and pink histograms means that the value of the blue histogram for the final bin - and therefore agreement or lack thereof with the experimental result - is not visible. Also, the significance of the values 3.25 bits and 3 bits in these plots should be discussed in connection with the input distributions.

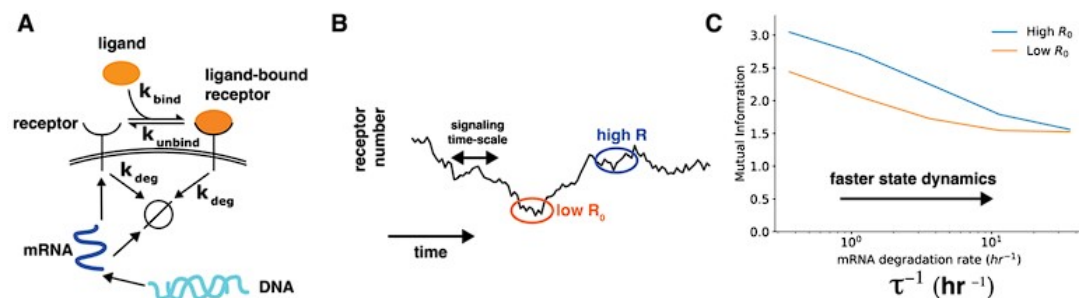
We thank the reviewer of the comment. We have now corrected our text (Figure 2 caption and lines 249-251).

1. While the joint distribution of the cell state dependent mutual information and various biochemical parameters is given in Fig. S7, there is no explanation of what these results mean, either in the SI or main text. Related to this, while a central claim of the work is that establishing this joint distribution will allow determination of cell state variables that differentiate between high and low fidelity sensing, this claim would be stronger with more discussion of Figs. 3 and S7. The related central claim that cell state dependent mutual information leads to higher fidelity sensing at the population level would be made stronger if it can be demonstrated that in the limit of rapidly varying cell state variables, the I_{CSA} is retrieved.

We thank the reviewer for this excellent comment. We have now added more discussion about interpreting the correlation between cell state variables and cell-state specific mutual information (lines 294-306). We also appreciate the suggestion about a toy model calculation to show that dynamics of cell state variables affects cell state specific mutual information. We have now performed a simple calculation to show how dynamics of cell state variables affects cells' sensing ability (lines 325-363). Specifically, we constructed a model of a receptor binding to the ligand wherein the receptor levels themselves changed over time through a slow process of gene expression (Author response image 3, main text Figure 4). In this model, the timescales of fluctuations of ligand-free receptors on the cell surface can be tuned by speeding up/slowing down the degradation rate of the corresponding mRNA while keeping the total amount of steady state mRNA constant. As shown in Author response image 3, the dependence of cell-specific mutual information on cell state variable diminishes when the time scale of change of cell state variables is fast.

Author response image 3.

Author response image 3. Cell state dynamics governs cell state conditioned mutual information. A. In a simple stochastic model, receptor mRNA is produced at a constant rate from the DNA and the translated into ligand-free receptors. The number of ligand-bound receptors after a short exposure to ligands is considered the output. B. A schematic showing dynamics of receptor numbers when mRNA dynamics are slower compared to signaling time scales. C. Conditioning on receptor numbers leads to differing abilities in sensing the environment when the time scale of mRNA dynamics τ is slow. In contrast, when the mRNA dynamics are fast (large τ^{-1}), conditioning on cell state variables does not lead to difference in sensing abilities.



Reviewer #1 (Recommendations For The Authors):

My major concerns are mainly conceptual, as described below. With proper attention to these concerns, I feel that this manuscript could be a good candidate for the eLife community.

Major concerns:

1. The manuscript convincingly demonstrates that cells good sensors after all, and that heterogeneity makes their input-output functions different from each other. This raises the question of what happens downstream of sensing. For single-celled organisms, where it may be natural to define behavioral consequences at the single-cell level, it may very well be relevant that single-cell information is high, even if cells respond differently to the environment. But for cells in multicellular organisms, like those studied here, I imagine that most behavioral consequences of sensing occur at the multicellular level. Thus, many cells' responses are combined into a larger response. Because their responses are different, their high-information individual responses may combine into a low-information collective response. In fact, one could argue that a decent indicator of the fidelity of this collective response is indeed the population-level information measure estimated in previous works. Thus, a fundamental question that the authors must address is: what is the ultimate utility of reliable, but heterogeneous, responses for a multicellular system? This question has an important bearing for the relevance of their findings.

We thank the reviewer for this thought-provoking comment. We agree that the fidelity with which cells sense their environment, especially those in multicellular organisms, may not always need to be very high. We speculate that when the biological function of a collection of cells can be expressed as an average over the response of individual cells; high-information but heterogeneous cells can be considered equivalent to low-information homogeneous cells. An example of such a function is population differentiation to maintain relative proportions of different cell types in a tissue or producing a certain amount of extracellular enzyme.

In contrast, we believe that when the biological function involves collective action, spatial patterning, or temporal memory, the difference between reliable but heterogeneous population and unreliable homogeneous population will become significant. We plan to explore this topic in future studies.

1. *The authors demonstrate that the agreement is good between their inference approach and the direct estimation of response distributions from single-cell time series data. In fact, the agreement is so good that it raises the question of why one would need the inference approach at all. Is it because single-cell time series data is not always available? Is that why the authors used it for one example and not the other? The validation is an asset, but I imagine that the inference approach is complicated and may make assumptions that are not always true. Thus, its utility and appropriate use must be clarified.*

We thank the reviewer for the comment. As the reviewer correctly pointed out, live cell imaging data is not always available and has limited scope. Specifically, optical resolution limits measurements of multiple targets. Moreover, typical live cell measurements measure total abundance or localization and not post-translational modification (phosphorylation, methylation, etc.) which are crucial to signaling dynamics. The most readily available single cell data such those measured using single cell RNA sequencing, immunofluorescence, or flow cytometry are necessarily snapshots. Therefore, computational models that can connect underlying signaling networks to snapshot data become essential when imputing single cell trajectories. In addition, the modeling also allows us to identify network parameters that correlate most strongly with cellular heterogeneity. We have now clarified this point in the manuscript (lines 366-380).

Minor comments:

1. *I would point out that the maximum values in the single-cell mutual information distributions (Fig 2D and E) correspond to $\log_2$ of the number of inputs levels, corresponding to perfect distinguishability of each of the equally-weighted input states. It is clear that many of the mutual information values cluster toward this maximum, and it would help readers to point out why.*

We thank the reviewer for the comment. We have now included a discussion about the skew in the distribution in the text (lines 251-260).

1. *Line 216 references Fig 2C for the EGF/EGFR pathway, but Fig 2C shows the FoxO pathway. In fact, I did not see a schematic of the EGF/EGFR pathway. It may be helpful to include one, and for completeness perhaps also one for the toy model, and organize the figures accordingly.*

We thank the reviewer for the comment. We did not include three separate schematics because the schematics of the EGF/EGFR model and the toy model are subsets of the schematic of the IGF/FoxO model. We have now clarified this point in the manuscript (Figure 2 caption).

Reviewer #2 (Recommendations For The Authors):

- *the simple model of Fig. 2A would gain from a small cartoon explaining the model and its parameters.*

We thank the reviewer for the comment. We did not include a schematic for the toy model as it is a subset of the schematic of the IGF/FoxO model. The schematic of the toy model is included in the supplementary information.

- *L should be called u, and B should be called x, to be consistent with the rest of the notations in the paper.*

We have decided to keep the notation originally presented in the manuscript.

- *legend of 2E and D should be clarified. "CC input dose" is cryptic. The x axis is the input dose, the y axis is its distribution at the argmax of I. CC is the max of I, not its argmax. Likewise "I" in the legend for the colors should not be used to describe the insets, which are input distributions.*

We have now changed this in the manuscript.

- *the data analysis of the IGF/FoxO pathway should be explained in the main text, not the SI. Otherwise it's impossible to understand how one arrives at, or how to interpret, figure 2E, which is central to the paper. For instance the fact that $p(x|u, \theta)$ is assumed to be Gaussian, and how the variance and mean are estimated from the actual data is very important to understand the significance of the results.*

While we have added more details in the manuscript in various places, for the sake of brevity and clarity, we have decided to keep the details of the calculations in the supplementary materials.

- *there's no Method's section. Most of the paper's theoretical work is hidden in the SI, while it should be described in the methods.*

We thank the review of the comment. However, we believe that adding a methods section will break the narrative of the paper. The methods are described in detail in the supplementary materials with sufficient detail to reproduce our results. Additionally, we also provide a link to the github page that has all scripts related to the manuscript.

PS: please submit a PDF of the SI for review, so that people can read it on any platform (as opposed to a word document, especially with equations)

We have now done this.

Reviewer #3 (Recommendations For The Authors):

1. *Subplots in Fig. 1, inset in Fig. 3 are not legible due to small font.*

We have now increased the font.

1. *Mean absolute error in Fig. S5 and relative error in related text should be clarified.*

We have now clarified this in the manuscript.

1. *Acronyms (MACO, MERIDIAN) should be defined.*

We have now made these changes.

References

1. Gregor T, Tank DW, Wieschaus EF, Bialek W. Probing the limits to positional information. *Cell*. 2007;130(1):153-64. doi: 10.1016/j.cell.2007.05.025. PubMed PMID: WOS:000248587000018.
2. Cohen-Saidon C, Cohen AA, Sigal A, Liron Y, Alon U. Dynamics and Variability of ERK2 Response to EGF in Individual Living Cells. *Mol Cell*. 2009;36(5):885-93. doi: 10.1016/j.molcel.2009.11.025. PubMed PMID: WOS:000272965400020.
3. Gross SM, Dane MA, Bucher E, Heiser LM. Individual Cells Can Resolve Variations in Stimulus Intensity along the IGF-PI3K-AKT Signaling Axis. *Cell Syst*. 2019;9(6):580-8 e4.
4. Loos C H, J. Mathematical modeling of variability in intracellular signaling. *Current Opinion in Systems Biology*. 2019;16:17-24.
5. Dixit PD, Lyashenko E, Niepel M, Vitkup D. Maximum Entropy Framework for Predictive Inference of Cell Population Heterogeneity and Responses in Signaling Networks. *Cell Syst*. 2020;10(2):204-12 e8.
6. Taniguchi Y, Choi PJ, Li GW, Chen H, Babu M, Hearn J, Emili A, Xie XS. Quantifying E. coli proteome and transcriptome with single-molecule sensitivity in single cells. *Science*. 2010;329(5991):533-8. doi: 10.1126/science.1188308. PubMed PMID: 20671182; PMCID: PMC2922915.